VIEWER PRODUCT

Integration Guide

For an engineer at a common data environment — Asite, Procore, Dalux, or your own — who has to make this viewer open your customers' documents.

Read section 1 before planning anything. All four exchanges an integration needs are built, you can run a working host application from our repository today, and a stock install now serves the embed document and carries the `frame-ancestors` allow-list to put your origins on. **What has not been exercised end to end is a real document opening inside a real host frame**, so treat section 4 as the contract and your first integration as the thing that proves it. This guide says which parts you can build on now and which are newer than anyone's production traffic.

Since the last issue, nothing in the protocol has changed — section 5 is the same contract. What changed is what you inherit when you embed it: section 7 now states our accessibility and localisation position in enough detail to answer a procurement questionnaire without overstating it, including the one functional accessibility gap that is still open.

Companion document: **Technical Architecture**, for how the thing works internally.

Contents

1. What you can build against today

HOST CDE	EXCHANGE	VIEWER PRODUCT
Their web interface where the viewer appears	**1. mount the viewer →** **✓ BUILT** **iframe + cde.viewer.v1**	/embed route no session, no auth guard
Their identity provider Entra / Okta / own	**2. identify the user →** **✓ BUILT** **host.init.identity — 3 untrusted fields**	viewer-core 37 files · 6 services · 12 components
Their document store SharePoint / S3 / Blob	**3. hand over the document →** **✓ BUILT** **host.init.document, or POST /api/conversions**	Conversion pipeline fetch → scan → render
Their markup store they stamp the author	**← 4. return what was drawn** **✓ BUILT** **3 markup events + 1 operation pair**	host-channel origin-checked in both directions
	⚠ NOT YET DEPLOYABLE **the allow-list exists; nothing serves the embed document**	

Figure 1 — All four exchanges now have code on both sides, checked against each other by a demo host written in plain JavaScript. The `frame-ancestors` allow-list the backend refused to have now exists — `cde.web.embed-parent-origins`, closed until a deployment names a host. What is still missing is a tier that serves the `/embed` document at all: the backend image carries no frontend, and the Kubernetes manifests route everything to the backend.

Two things are true at once, and conflating them will cost you a sprint.

The protocol is real. `cde.viewer.v1` is implemented on both sides, and our repository ships a host application — plain HTML and JavaScript, no framework, no build step — that frames the viewer, drives the handshake, stores markup, and refuses an operation. You can clone it, run it, and read it as the thing you are about to write. Section 5 tells you how.

The deployment now exists, and is new. The image serves the viewer's pages itself when `cde.web.app.path` points at a staged build, so one container answers both `/embed` and the API, and emits the `Content-Security-Policy` that governs the document it just served. `cde.web.embed-parent-origins` names the origins permitted to frame that route and is closed until someone sets it — so ask your deployment contact to add your origin before you test, because until they do the frame comes up blank with an error only your browser console will show you.

What that deployment has been through. The demo host in our repository has been run against it end to end: the document opens, pages render, markup is drawn and stored by the host, the host page is reloaded and the markup comes back and is painted, and the host drives page and zoom. That run found six defects, all fixed, none of which any unit test in this repository had caught. What it has *not* been through is a real CDE, a real document store and real users — so treat section 4 as the contract and budget a spike for your first integration.

The conversion API has no such caveat. Section 3 is a complete, working integration you can ship against today, inside your own interface, with your own rendering — and for a "preview any file format" feature that is often the whole requirement.

2. The shape of it

Your CDE keeps its documents and its users. The viewer borrows a document briefly, converts it, and discards it.

We never hold your credentials and never learn which storage platform your link points at. SharePoint, S3, Azure Blob and GCS all reduce to "a URL that works for fifteen minutes", which is the property that makes this portable across CDEs rather than a per-vendor connector.

3. Converting a document from your storage

#	STAGE	WHAT HAPPENS
1	**Host mints a link**	Short-lived, single-object, your credentials. We never hold them.
2	**Submit**	POST /api/conversions { sourceUrl, targetFormat } — 202 + Location header, under one second
3	**Destination policy**	Resolves the hostname and validates the ADDRESS, not the string. Refuses loopback, link-local, 169.254.169.254, RFC 1918, .internal. Redirects are NOT followed.
4	**Bounded fetch**	Size cap enforced during the stream, not after. Connect and read timeouts.
5	**Quarantine + scan**	Magic-byte type check, then ClamAV. Not referenced by the application until clean.
6	**Convert**	Sandboxed, resource-limited, network-isolated worker, out of process.
7	**Poll**	GET /api/conversions/{jobId} — PENDING, RUNNING, then SUCCEEDED / FAILED / CANCELLED (terminal)
8	**Collect**	GET /api/conversions/{jobId}/content — streamed

Figure 2 — The conversion API, end to end. Stages 1, 7 and 8 are yours; 2 to 6 are ours. Always send an Idempotency-Key so a timed-out retry returns the same job rather than converting twice. An embedded viewer opening a PDF skips this path entirely — the browser fetches your URL directly and no server of ours sees the file.

3.1 Mint the link

Generate a short-lived, single-object download URL with your own credentials — a Microsoft Graph download URL, an S3 presigned GET, an Azure blob SAS, a GCS signed URL. Keep the expiry tight; we fetch once, immediately.

Do **not** send us a permanent URL, a URL that needs a credential we would have to store, or a path on your internal network — section 3.5 explains what happens.

3.2 Submit

```
curl -X POST https://viewer.example.com/api/conversions \
  -H "Authorization: Bearer $TOKEN" \
```

```
  -H "Content-Type: application/json" \
  -H "Idempotency-Key: 8f14e45f-ceea-467a-9f3a-1d2c9b7e4a51" \
  -d '{
        "sourceUrl": "https://files.example.test/drawings/site-plan.dwg?token=synthetic",
        "targetFormat": "PDF"
      }'

202 Accepted
{ "jobId": "3f2a71c4-9b0e-4a2d-8c11-5e7d9a1b3f40", "status": "PENDING",
  "sourceHost": "files.example.test", "targetFormat": "PDF" }
```

Both fields are required. `targetFormat` accepts `PDF` and nothing else today — it is in the contract so that adding a second output is additive rather than breaking. The `jobId` is a UUID; use it verbatim on every other endpoint.

Requires the `document:convert` permission. Returns in under a second — the work has not happened yet, and that is the design: an endpoint either answers in under a second or hands back a job id in under a second.

> **Always send `Idempotency-Key`.** A timed-out retry then returns the same job rather than converting the same file twice, and conversion is the expensive operation in this system.

Note what is not echoed back: the URL. Only its host is kept on the job record. The link is a bearer credential, so it is never written to the database, a log, or a message — which also means you cannot read it back to find out what you sent.

3.3 Poll

```
curl https://viewer.example.com/api/conversions/3f2a71c4-9b0e-4a2d-8c11-5e7d9a1b3f40 \
  -H "Authorization: Bearer $TOKEN"

200 OK
{ "jobId": "3f2a71c4-9b0e-4a2d-8c11-5e7d9a1b3f40", "status": "SUCCEEDED",
  "sourceHost": "files.example.test", "sourceFileName": "site-plan.dwg",
  "targetFormat": "PDF", "resultSizeBytes": 486213, "failureReason": null }
```

`PENDING` while queued, `RUNNING` while working, then one of `SUCCEEDED`, `FAILED` or `CANCELLED`. **Those three are terminal and never change again**, so stop polling when you see one. On `FAILED`, read `failureReason`.

There is no progress percentage and no sub-stage — the job reports the state it is in, not how far through it is. Back off as you poll; a large IFC will not finish in the first second, and hammering the endpoint only costs you rate limit.

3.4 Collect

```
curl https://viewer.example.com/api/conversions/3f2a71c4-9b0e-4a2d-8c11-5e7d9a1b3f40/content \
  -H "Authorization: Bearer $TOKEN" -o converted.pdf
```

Streamed, with `Content-Disposition: attachment` and `X-Content-Type-Options: nosniff`. Cancel an in-flight job with `DELETE /api/conversions/{jobId}`.

3.5 What we refuse, and why

Your URL is dereferenced by our server, which makes it an SSRF vector. Every submission goes through a destination policy that:

- **resolves the hostname and validates the resolved address**, not the string — a name that resolves to `169.254.169.254` is refused however it is spelled;
- refuses loopback, link-local, RFC 1918 (`10/8`, `172.16/12`, `192.168/16`), `::1`, and `.internal` names;

- **does not follow redirects at all.** Not “follows a capped number and re-validates” — none. Your link must resolve to the bytes directly;
- requires `https` unless the deployment has explicitly opted out;
- enforces the deployment’s list of permitted storage hosts, where one is configured;
- bounds the fetch: size enforced **during** the stream, plus connect and read timeouts.

> **The redirect rule is the one most likely to break you.** Plenty of storage 302s to a CDN. Test the exact URL you intend to send, not a URL that eventually reaches the bytes. And ask which hosts are allow-listed on the deployment before you integrate.

After the fetch: content type is decided by **magic bytes**, never by your `Content-Type` header or the file extension; the file lands in a quarantine prefix; ClamAV scans it; only then does it convert. An infected file is deleted and the event audited.

3.6 Errors

RFC 9457 problem documents throughout, with a `traceId` you can quote.

```
{ "type": "/problems/fetch-not-permitted",
  "title": "Fetch not permitted",
  "status": 422,
  "detail": "The source URL resolves to a private address (10.0.4.19).",
  "traceId": "4f8a1c2e9b7d6a5f3e2d1c0b9a8f7e6d" }
```

Endpoint	Status	Means
POST /api/conversions	**202**	Accepted. The Location header names the job to poll — use it rather than assembling the URL yourself
	422	The link may not be fetched: wrong scheme, an unlisted host, or a refused destination. This is the SSRF refusal — it is a 422, not a 403
	429	The conversion queue is full, not a per-caller rate limit. Retry after the interval in Retry-After
GET /api/conversions/{jobId}	**404**	No such job is visible to you. Note the wording: another tenant’s job is invisible, not forbidden
DELETE /api/conversions/{jobId}	**404**	As above
GET /{jobId}/content	**409**	The job has not succeeded, so there is nothing to download yet. Poll to a terminal state first

Table 1 — Status codes you will actually meet.

Read `detail` and show it. It is written to be shown. Quote `traceId` to support — it is the only thing that ties your failure to our logs.

4. Formats

Input	Status	Route	Notes
PDF	**Supported**	Passed through; pdfjs-dist	OCR available for scanned pages, adds an invisible text layer
DXF	**Supported**	ezdxf → SVG or PDF	Text is real and searchable in both

Input	Status	Route	Notes
Office	**Supported**	LibreOffice → PDF	docx, xlsx, pptx
IFC	**Supported**	Geometry + hierarchy tree	Tree is the primary interface, not a fallback
DWG	**Needs ODA**	ODA → DXF, then as above	The image ships no DWG reader. The operator supplies an ODA File Converter — see below
DWF	**Not supported**	—	No route exists. Autodesk's Design Web Format is not read by any component of the pipeline
RVT / RFA	**Not supported**	Detected and refused	Recognised by OLE2 magic bytes and rejected with REVIT_BINARY. Identified only so the error is clean

Table 2 — What the pipeline actually accepts. Four supported outright, one conditional, two not supported at all.

DWG now requires an ODA File Converter that the operator supplies. This changed, and in your favour if you were reading the previous edition. That edition said DWG worked in our hosted service and was unshippable on-premises, because the image bundled LibreDWG (GPL-3.0) and distributing it owed recipients corresponding source we did not provide. **LibreDWG has been removed.** Nothing encumbered is distributed, and the same rule now applies everywhere: mount an ODA installation, point `ODA_PATH` at it, and DWG works — hosted or on-premises, no distinction. Without one, the deployment has no DWG reader at all. ODA is licensed by you or your operator directly from the Open Design Alliance; check `odaRunnable` rather than `odaInstalled` in the status response, because a mount missing its libraries or its execute bit reports as present and is useless.

Only PDF opens inside an embedded frame today. Everything in this table is supported through the conversion API in section 3. The embed reaches the browser-rendered path only, so an embedded viewer asked for a `.docx` or an `.ifc` returns a 415 naming the format and saying the conversion service is what is missing. If your embedded use case is not PDF-first, raise it — the work is wiring, not design.

4.1 Why RVT and DWF are absent, and what it would take

These two are named separately because they are the ones most often assumed, and because the reason they are missing is not "nobody got to it yet".

RVT (and RFA) is a proprietary Autodesk binary with no open reader. The pipeline detects it by OLE2 magic bytes and refuses it deliberately, so you get a clean `REVIT_BINARY` error rather than a corrupt render. Every route to supporting it is commercial.

Route	What it costs
Autodesk Platform Services (formerly Forge)	A cloud API. It SENDS THE MODEL TO AUTODESK, which collides directly with the data-residency and sovereignty position — for a Defence or IRAP-scoped tenant it is not merely a cost question, it is prohibited
Revit itself, plus an export plugin	Windows hosts, a licence per seat, and a rendering farm that is not the product's architecture
A commercial SDK (ODA BimRv or equivalent)	A vendor relationship and a per-deployment fee. ADR 13 settled the same question for DWG by making ODA operator-supplied rather than bundled, so the shape of the answer is

Route	What it costs
	known — it is a commercial conversation, not an open one

Table 3 — The three routes to RVT, and why none is free.

> The sovereignty collision is the important one. **The cheapest route is the one we can least use**, because the tenants most likely to hold Revit models are also the ones whose data must not leave the jurisdiction.

DWF is unimplemented rather than refused. It is a ZIP container holding W2D and W3D streams, and reading it is ordinary engineering rather than a licence problem — but no component of the pipeline knows the format today, so a DWF submission will fail as an unrecognised type. If DWF matters to you, say so: it is schedulable work in a way RVT is not.

If you need either, raise it as a commercial requirement, not a bug. The answer involves a vendor contract and, for RVT, a residency decision that engineering cannot take alone.

5. Embedding — built, and what your deployment must configure

An iframe and a versioned postMessage protocol. The viewer authenticates nobody and authorises nothing: you mint a short-lived URL for the document, tell it a name to display, and receive what the user did — every operation that changes anything comes back to you to authorise server-side.

Recorded as ADR 14 (`cde-platform`, `docs/adr/0014-viewer-embed-and-identity-contracts.md`), with the message-level detail in `docs/viewer-embed-protocol.md`. Both sides are now implemented, and the protocol document is the normative one — where it and this guide disagree, it wins.

5.1 Run the demo host first

`demo/` in the frontend repository is a host application that frames the viewer. It is not the viewer, and that is the point: **it is the side you are about to write**, in plain HTML, CSS and JavaScript with no framework, no build step and no dependencies, because your stack is your business.

```
npm start                # the viewer, on :4200
node demo/server.mjs     # the host,   on :4401

# then open http://localhost:4401
# viewer somewhere else?
VIEWER_ORIGIN=https://viewer.example node demo/server.mjs
```

Two ports deliberately: a different port is a different origin, so the demo exercises the real cross-origin path — origin checks on every message, CORS on the document fetch, `frame-ancestors` on the viewer. A demo served from one origin would pass with every one of those broken.

It ships three generated sample documents and a message log down the side, so the handshake is visible as it happens. Worth trying in this order: open the drawing and watch `viewer.ready` → `host.init` → `viewer.loaded`; draw on it and see the markup appear in the host's store; reload the page and reopen the document, and watch the markup come back — the viewer forgot, the host remembered; change the display name and draw again, and see the new name stamped on a markup the viewer never put a name on. Then tick

`document:sign` and use Sign: the control renders and the operation is **refused**, and both of those are correct.

5.2 The one thing your viewer deployment must change

> **A deployment that has not been told about you refuses your iframe, and the failure looks like a blank frame rather than an error.** `frame-ancestors` is `'none'` until your origins are named, deliberately: opening the embed is a configuration act and never a default. This is not something you can set from your side.

The setting is `cde.web.embed-parent-origins` — a list of exact origins, `'none'` when empty, and validated at startup. Wildcards, the `null` origin, CSP keywords and anything carrying a path are all refused by name, because **a `frame-ancestors` source a browser cannot parse is not a closed door**: the browser drops what it cannot read and applies the rest, so a typo widens the policy rather than breaking visibly. Give your deployment contact the exact origins you will frame from, including scheme and any non-default port.

This header is the **authorisation** decision about who may frame the viewer. The `parentOrigin` you configure in the handshake is only addressing — it says where the viewer should post, not who is permitted to frame it. A deployment that relaxed only the second would be framable by anyone who sent the right message.

> **Ask your deployment contact which component serves `/embed`.** The setting governs what the *backend* answers. A stock install now serves the embed document from the backend image itself, so one component emits the document and the header together and there is nothing else to configure. But a deployment that puts the Angular build behind its own web tier has that tier serving the document, and **it needs the same `frame-ancestors` value** — the backend's copy governs only what the backend answers. Two questions, not one: which of your origins are on the allow-list, and what serves the document.

5.3 The minimum integration

Four steps, and none of them involve a token.

- **Frame the embed route** and listen for messages, checking `event.origin` against the viewer origin on every single one — not once at setup.
- **Wait for `viewer.ready`**, then post `host.init` with the document (a short-lived URL, its media type, a display name, and your own `externalId`) and, if the user may do more than read, an identity block.
- **Store what comes back.** `viewer.markupCreated`, `…Updated` and `…Deleted` carry the markup; `shapeData` is an opaque string you store and hand back, never parse. Stamp the author yourself from the session you already hold.
- **Listen to as much of the rest as you need, and none of it if you do not.** Section 5.4 lists the lifecycle events. Every one is optional.
- **Answer `viewer.operationRequest`** with `host.operationResult` — `applied`, `refused` or `failed`. One message pair covers all seventeen document operations; they are not seventeen message types.

> **Send no credential in any message, in either direction.** The document URL is a bearer credential with a short life and is the only thing resembling one that crosses the boundary; it is never stored, logged, or echoed back to you. Never post to `'*'`.

One subtlety worth internalising early: **a `refused` must be possible even for an operation whose capability you granted in `host.init`.** Capabilities decide which controls render; authorisation happens when the operation is requested. If the two ever

disagree — the user's permission changed thirty seconds ago — the `refused` is correct and the rendered button was merely stale.

Three operations never become viewer-side, whatever else changes: `document.sign`, `version.create` and `version.restore`. A signature over a copy you have since replaced is worse than no signature, and the viewer cannot know whether its copy is still current.

One constraint from the previous edition of this guide is unchanged and still catches people: **the CORS source registers no origin at all unless the deployment names one in full.** For a PDF, the browser fetches your URL directly, so your storage must allow the viewer's origin to read it.

5.4 Events you can hook, and what each is honestly for

Thirteen message types run viewer → host. Four exist purely so you can **record** what a reader did without polling us or inferring it from markup traffic — and the inference is wrong in ways that are not obvious, which is why they exist.

> The last issue of this guide said fourteen, and it was counting one that never arrives. `viewer.resized` was specified, listed in the protocol's own event table with a payload shape, and sent by no code path — so a host that sized its frame to content waited for an event that never came, with nothing erroring to say why. It has been removed from the protocol rather than left as a promise, and a CI gate now refuses any declared message type that nothing sends. If you want the frame to follow the content, measure it from your own page.

You want to know	Listen to	What to watch out for
A document was opened	viewer.opened	Fires before the viewer knows whether it can render the file, and before any fetch. Exactly one of viewer.loaded or viewer.error follows it
It rendered	viewer.loaded	Carries pageCount, mediaType and what rendered it
Your stored markup arrived	viewer.markupLoaded	Check `rejected`. Non-zero means your store holds markup the viewer cannot draw — a data problem on your side that was previously silent
The user drew, changed or removed something	viewer.markupCreated / …Updated / …Deleted	No author field. You stamp it from the session you already hold (5.5)
The user selected a markup	viewer.selectionChanged	markupId, or null when nothing is selected
Which pages were actually read	viewer.pageRendered	Once per page per document. Zooming repaints a page and does NOT re-announce it, which is what makes a "pages read" count mean anything
Where the user is now	viewer.viewChanged	Throttled to 4/s. Use this to follow a reader live; use pageRendered to record what was seen
A document stopped being shown	viewer.unloaded	Carries the id of the document that CLOSED, not the one arriving. Not guaranteed on teardown — see below

Table 4 — The events worth hooking, and the trap in each.

> **Do not treat `viewer.unloaded` as a guarantee.** If your page removes the iframe, navigates away, or the tab closes, nothing can post from a frame that no longer exists. Use it to close a record you are already keeping, never as the only place you write one — the same caveat that applies to `beforeunload` in your own page, for the same reason.

Two more sharp edges worth knowing before you build against these. A document the viewer **rejects outright** — a descriptor missing `mediaType`, say — produces `viewer.error` with no `viewer.opened` before it, because there was no document to open. And a `host.loadMarkup` whose `markup` is not an array is **ignored silently and acknowledges nothing**: if you get no `viewer.markupLoaded`, check you sent an array.

All four were added after the protocol shipped, which is the compatibility promise working as intended — new message types are additive within v1, so an integration written before them keeps running and simply never registers a handler. Full payloads: §6.3 of `docs/viewer-embed-protocol.md`.

5.5 Identity: there is nothing to configure

Not “you need to set up SSO”. The embed path has no identity system to configure at all. Our own `/api/annotations` still keys a markup’s author to a row in our user table — integrating through *that* API would mean shadow-provisioning your users into our database, a data-protection conversation before it is an engineering one. The embed never reaches it.

THE PLATFORM’S OWN API — unchanged	THE EMBED PATH — built
Priya, at Asite host identity	Priya, at Asite known only to Asite
↓ **blocked**	↓
Annotation.author @ManyToOne User	host.init.identity display name · subjectId · capabilities
↓	↓
OUR users table she must exist as a row	Nothing persisted the viewer stamps no author
Integrating through this API means shadow-provisioning every host user into our database.	The host stays the system of record, and stamps the author itself when it stores the markup.

Figure 3 — Both paths exist in the code. `/api/annotations` still keys an author to a row in our own user table, and always will for this platform’s own users. The embed path never reaches it: the viewer emits a markup with no author at all, and the host attributes it from the session it already holds.

Three untrusted fields, and that is the whole of it. You send a display name, an opaque subject id, and a list of capabilities. All three are used for presentation and for deciding which controls render; none is trusted for authorisation, because the browser is not a place authorisation happens. The author is stamped by whoever persists the markup — you. Nothing about your users reaches our database.

> You do **not** need to mint a JWT, and you do not need an identity provider we can talk to. If an earlier version of this guide told you to prepare a signed assertion with a stable subject claim, disregard it — that was written while the question was open, and the answer went the other way. **Sending no identity at all is a supported read-only deployment**, not a degraded one.

One consequence to plan around: **live collaboration is not available in an embed.** Cursors and presence ride a socket authenticated by a viewer session, and an embedded viewer has no session by design. That is the acknowledged cost of “the viewer authenticates nobody”, and it is unsolved rather than decided against.

6. What is settled, and what is still open

Everything in the first list is decided and implemented — design against it. Everything in the second is a real gap we would rather you heard from us than discovered.

Settled

- **Embedding is an iframe** with a per-tenant `frame-ancestors` allow-list. Not a web component: the viewer renders untrusted documents, and sharing your origin would mean a document that escapes the PDF renderer runs with your session on your domain.
- **Identity is three untrusted fields** — a display name, an opaque subject id, and capability flags that decide which controls render and nothing else. No signed assertion, no viewer session, no JWT.
- **Markup comes back as events**, with the geometry as an opaque string you store and hand back. There is no author field — you stamp that.
- **Document identity is yours.** Pass an `externalId` in the handshake and it comes back on every event, so you never hold a map between your id and ours.
- **One message pair covers all seventeen document operations**, and three of them — sign, create version, restore version — will never move to the viewer.
- **The protocol version is in every message, and `viewer.ready` tells you which versions the deployment speaks.** Read it and pick; do not assume. Within v1 we may add message types, optional fields, commands and operations — you ignore what you do not recognise, which is why the rule is *validate*, not *reject on unknown field*. Removing anything needs v2, announced with at least six months’ notice and both versions running over the overlap.

Still open

- **No real CDE has integrated it yet** — section 5.2. Our own demo host drives the whole protocol against a served viewer; your storage, your documents and your users are the parts nobody has exercised.
- **Non-PDF formats in an embedded frame** — section 4. Wiring, not design.
- **Collaboration in an embed** — section 5.5. Genuinely unsolved.
- **Accessibility evidence.** See section 7 before you rely on ours.

Full message shapes: `docs/viewer-embed-protocol.md`. Where this guide and that document disagree, that document is the normative one.

7. Obligations you inherit

Accessibility. WCAG 2.2 AA is a procurement gate in the UK, EU, Australia and the US, and an embedded viewer becomes part of *your* conformance claim. If you embed our WebGL model view, you inherit the requirement for an equivalent accessible route to the same information — we provide the hierarchy tree for exactly this reason, and it needs to remain reachable in your integration. Exports must be tagged and PDF/UA-conformant.

Do not inherit a conformance claim from us, because we are not making one yet. An accessibility statement, a VPAT 2.5 INT conformance report and a screen-reader matrix all exist in our repository, and all three record the same thing: no criterion has been evaluated by an audit. If your bid depends on ours, ask for the current state in writing rather than citing this guide.

What we can tell you honestly, because each item has a test that fails when it regresses:

- The model hierarchy tree implements the full tree pattern — roving tab stop, arrow navigation, `aria-level` and `aria-expanded` per row.
- Page reordering and the comparison wipe, both previously drag-only, have keyboard routes (SC 2.5.7).
- Icon-only controls carry accessible names rather than announcing their glyph.
- Form controls are programmatically associated with their labels, their validation messages and their hints; required fields say so to assistive technology rather than only showing an asterisk.
- Long operations and their results are announced through live regions.
- **Every route now names itself with a top-level heading** (SC 2.4.6). Four did not, including the document viewer itself, so a reader navigating by headings reached the viewer and found nothing naming the open document. Where a visible title already existed it carries the heading semantics now rather than gaining a hidden duplicate; where none existed the heading is hidden, because a second visible title on a sign-in card is chrome nobody asked for. The embedded frame carries its own, since an iframe is a separate document and your `h1` is not inside it — hidden, so nothing we inject appears in your page.
- The embedded toolbar is marked as a group of named buttons rather than a toolbar. It used to claim `role="toolbar"`, which promises arrow-key navigation over a single tab stop that the row does not implement — so a screen-reader user was told it was a toolbar, pressed the arrows, and nothing moved. If your own audit checks ARIA roles against their behaviour, this is one fewer finding than the last release would have given you.

And what we cannot. No axe run, no Lighthouse budget, no screen-reader pass against the supported matrix, and no CI gate producing any of them. If your own audit runs axe, expect it to find things this document does not list: the headings defect above is exactly the kind a tool catches and we did not, and it sat in our own conformance report mis-described for a release.

The heading skip the last issue of this guide reported is fixed, and the reason it gave for leaving it was wrong. That issue said levels skipped from `h1` to `h3` in the viewer, compare and 3D routes and that fixing it needed the route outlines designing, because the panels owning the `h3` appeared at more than one depth. They do not — each is used in exactly one place, which one check would have shown. The three routes now run `h1` to `h2` to `h3` without a gap, and in the viewer the new `h2` also names the sidebar panel you are in, which seven of the ten panels previously left unnamed.

The labels half of SC 2.4.6 has now been audited too, which the last issue of this guide listed as outstanding. It found eight controls with no accessible name and all eight are fixed: both document searches and three form-field controls had only a placeholder, which disappears the moment a reader types; the viewer’s panel toggle announced as nothing; the per-document status chooser announced the same name on every card; and **every link annotation in every PDF was an empty link** — a page with fifty cross-references put fifty nameless links in the tab order. A CI gate now fails the build on a control with no name.

Two limits on that, because it should not be read as more than it is.

- **The gate checks that a name exists, not that it is apt.** Whether a label describes its control well is a judgement no tool makes.
- **It cannot verify a label whose `for` and a control whose `id` are both bound to expressions** — correct code that no static reader can pair up. A genuine mismatch of that shape passes. Catching it needs the rendered page, which is what the manual pass and the screen-reader matrix are for, and why neither is a formality.

The SC 2.5.7 failure in the annotation layer that the last issue of this guide reported is **fixed for drawing**: a shape can now be created entirely from the keyboard on both the PDF page and the converted drawing. Focus the markup layer, then arrow keys move a crosshair, Shift takes a larger step, Space presses, Enter finishes and Escape cancels — the arrow keys between the press and the finish are the single-pointer alternative to the drag.

Two limits on that, stated plainly because a bid should not over-claim.

- **A committed shape still cannot be moved or resized — by anyone.** The viewer has no shape selection and no move, for any input device, so a misplaced annotation is deleted and drawn again. The last issue filed this as half of the keyboard gap, which hid it; it is a functional gap, not an accessibility one.
- **A reader cannot yet enumerate existing markup** through assistive technology. Creating is covered; reading back what is on the page is not.

> If your customers annotate drawings and your bid claims AA, raise both with us before you sign.

Localisation. Everything a reader sees carries a stable message id and a translator note, extracted to a committed catalogue — 595 messages — with two CI gates keeping it honest: one fails if the catalogue drifts from source, the other if a template grows text a translator will never see.

> **The last issue of this guide said there were no hardcoded strings left. Nine were found after it shipped,** and six of them were in the embedded viewer — the surface *your* application frames, so the English appeared inside your translated interface. They are fixed. They are mentioned because of what they say about the gates rather than about the nine: all nine sat where neither gate looks, in expression operands and return statements rather than in template text or a `$localize` call. The catalogue is complete as far as two mechanical sweeps can see, and a string placed where neither looks will still get through. If you are selling into a non-English market, spot-check the embedded surface in your target locale rather than taking the catalogue's completeness on trust.
>
> **We ship the source catalogue, not translations.** The messages are English and there are no other locales in the repository. If you sell into a market that needs French or Arabic, you supply the translated catalogue; the mechanism to consume it is Angular's standard `$localize` pipeline and costs you a build per locale, not a fork. RTL layout has been kept in mind throughout — logical properties rather than `left`/`right` — but has not been verified against a real RTL locale, because there is not one to verify against.

A few sentences still reach the screen in English from the server, in the places where only the server knows what happened. We have replaced them wherever the response also carries a status code the client can word itself, which is most of them, but a host serving a non-English market should expect a small residue and ask us to name it for the paths they care about.

Trademarks. "Works with Microsoft SharePoint" is nominative fair use. "Microsoft-approved", their logo, or any implication of partnership is not. The same applies to Amazon, Google and Autodesk. Integration documentation is where this goes wrong most easily, in both directions.

Certification claims. Do not describe this product as SOC 2, ISO 27001 or IRAP certified. Those certifications are not currently held, and stating otherwise is a misrepresentation with regulatory consequences rather than a marketing stretch.

8. Integration checklist

Today

- Can you mint a short-lived, single-object download URL from your storage?
- Does that URL resolve to the bytes directly, with no redirect?
- Do you have somewhere to hold a job id between submit and collect?
- Do you generate an `Idempotency-Key` per logical submission?
- Do you back off while polling, and honour `Retry-After` on `429`?
- Do you surface `detail` and `traceId` from problem documents to your support path?
- Have you confirmed your DWG position?

Before you scope an embedded integration

- Have you run the demo host and watched a handshake? (`demo/README.md`)
- Have you read the embed protocol? (`docs/viewer-embed-protocol.md`)
- **Have you given your deployment contact the exact origins you will frame from, and confirmed they are on `cde.web.embed-parent-origins`?**
- **Have you confirmed what will serve the `/embed` document on that deployment?**
- Can you mint a short-lived URL for a document, and serve a page that frames us?
- Does your storage allow the viewer's origin to read that URL from the browser?
- Do you check `event.origin` on every message, not once at setup?
- Do you have somewhere to store markup that is not the rendered file, and do you stamp the author from your own session rather than from the message?
- Do you know which of the 17 document operations you need to implement as callbacks, and can you return `refused` as readily as `applied`?
- Is your embedded use case PDF-first, or do you need the conversion path wired in?

The integration you can ship against a stock deployment today is still section 3. The embed is built and demonstrable, and it needs one header changed on the viewer side before it runs anywhere but a development install. Scope it as something to plan with us rather than something to build against unilaterally — and if all you need is viewing, converting through section 3 and rendering the result yourself remains a complete feature with none of these caveats.